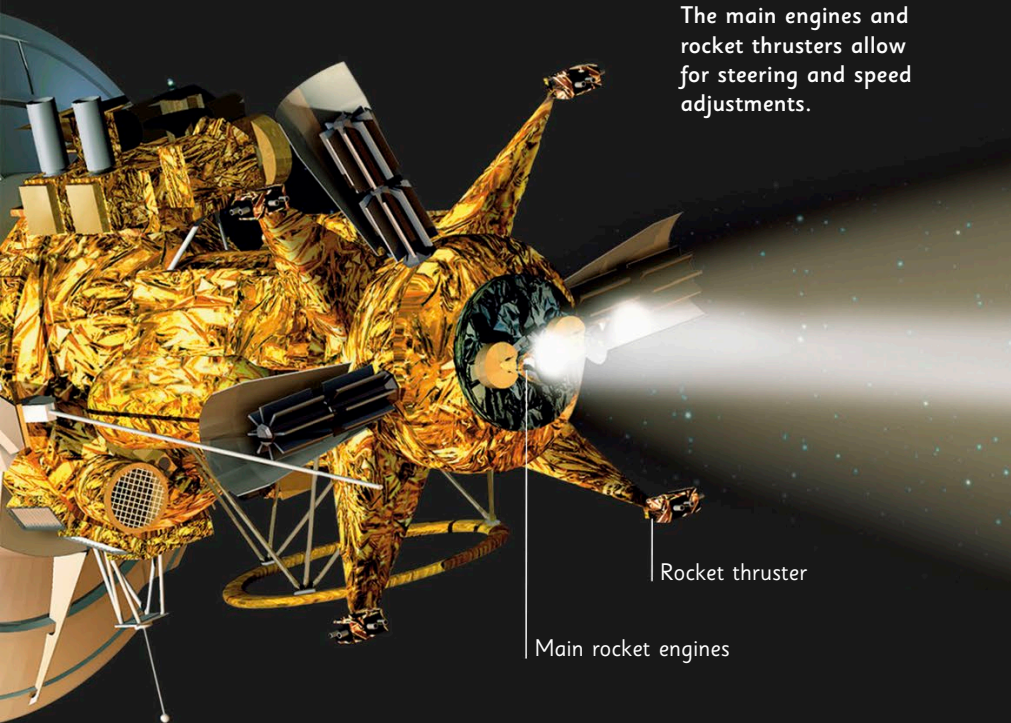


As the crow flies?

Spacecraft launched from the Earth do not necessarily travel to their destination in a straight line. The *Cassini-Huygens* orbiter and probe, for example, took a roundabout route during its mission to Saturn in order to make use of gravity assists (see below).

Cassini orbiter



The main engines and rocket thrusters allow for steering and speed adjustments.

Rocket thruster

Main rocket engines



What is a gravity assist?

By flying past a planet and making use of its gravity to change speed and direction, a gravity assist helps spacecraft on their way. *Cassini-Huygens* travelled twice past Venus, once past the Earth, and once past Jupiter before it headed for Saturn.

Picture detective

Look through the Exploring Space section and see if you can identify the pictures below.



Turn and learn

Rockets:
pp. 30-31
Space stations:
pp. 38-39